

	SGGS Institute of Engineering and Technology, Vishnupuri, Nanded	
	ISE-II- Examination (Semester-I) 2025-26	
	Department : Chemical Engineering	Class :Final Year
	Sub: Waste Water Treatment	Sub. Code: PCC-CH418
Date: 10 /10/2025	Time: 14.30-15.15	Max. Marks :20

- Note: i. Attempt any four questions
ii. Assume suitable data if necessary

Q.N	Questions	Marks	CO	BT
Q 1	What are the sources of water pollution? Explain in brief.	05	1	1&3
Q 2	What do you understand by precipitation process? Describe chemical precipitation in wastewater treatment.	05	2&3	2&3
Q 3	What do you understand by COD? Differentiate between BOD and COD in detail.	05	1&2	2&4
Q 4	What is aerobic digestion, and how does it differ from anaerobic digestion?	05	1&3	1&4
Q 5	What is the purpose of screening in wastewater treatment? Explain in brief with neat sketch	05	1&3	1&3

Course Outcomes (CO)

1. Understand the water standards and disposal norms
2. Analyze waste water samples for several parameters.
3. Utilize various unit operations for waste water treatment.
4. Designing of waste water treatment plant.
5. Treat waste water with advanced processes

Bloom's Taxonomy (BT)

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Chemical Engineering
In-Semester Examination II
Academic Year: 2025-26 (semester I)

Class: Final Year B. Tech.

Course: Sustainable Chemical Technology

Course Code:
PEC- CH409

Date: 09-Oct-2025

Time: 14:30 to 15:15

Max. Marks: 20

Note:

- i. *All questions are compulsory.*
- ii. *Assume suitable data whenever necessary and state it clearly.*
- iii. *Figures to right indicate full marks.*
- iv. *The symbols carry usual meaning.*

Q.N.			Marks	CO	BT
Q.1	a	Discuss in detail the life cycle analysis (LCA)?	4	CO3	BT-VI
	b	Explain environmental impact assessment (EIA)?	4	CO3	BT-II
	c	What are the basic concepts of conventional and non-conventional energy resources?	4	CO3	BT-I
Q.2	a	Define industrial ecology and industrial symbiosis?	4	CO4	BT-I
	b	Explain the ISO 14001:2015 framework and discuss its benefits?	4	CO4	BT-II

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Understand the different types of environmental pollution problems and their sustainable solutions.
CO 2	Environmental Pollution: Air Pollution and its effects, Water pollution and its sources, zero waste concept and 3 R concepts in solid waste management.
CO 3	Environmental management standards: ISO 14001:2015 framework and benefits, Scope and goal of Life Cycle Analysis (LCA),
CO 4	Environmental Impact Assessment Studies Sustainable Habitat
CO 5	Basic concept of sustainable habitat, Methods for increasing energy efficiency in buildings.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Chemical Engineering
In-Semester Examination II
Academic Year: 2025-26 (semester I)

Class: Final Year B.Tech.	Course: Catalytic Science and Engineering	Course Code: PEC-CH408
Date: 09.10.2025	Time: 14:30 to 15:15	Max. Marks: 20

Note:

- All questions are compulsory.
- Assume suitable data whenever necessary and state it clearly.
- Figures to the right indicate full marks.

Q.N.	Question	Marks	CO	BT
Q. 1	Describe the construction and working of UV-Visible spectroscopy.	5	3	2
Q. 2	Differentiate between scanning electron microscopy (SEM) and transmission electron microscopy (TEM).	5	3	4
Q. 3	Explain the construction, working principle, advantages, and disadvantages of a fluidized bed reactor (FBR).	5	4	2&3
Q. 4	Write a short note on the slurry reactor. Under what process conditions is a slurry reactor preferred over other types of reactors?	5	4	3&4

=== ALL THE BEST ===

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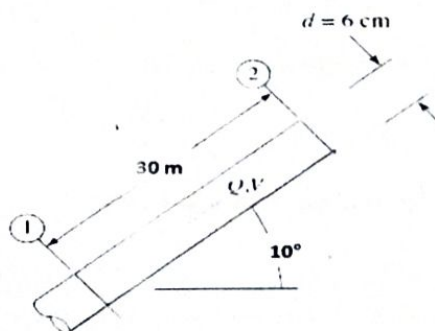
Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Apply knowledge of heterogeneous catalytic reactions in industrial applications.
CO 2	Find out the mechanism and kinetics of heterogeneous catalytic reactions
CO 3	Prepare the required catalyst and carry out characterization of the same
CO 4	Predict mass and heat transfer effects on heterogeneous catalytic reactions
CO 5	Design reactors for heterogeneous catalytic reactions.

	SGGS Institute of Engineering and Technology, Vishnupuri, Nanded		
	ISE-II- Examination (Semester-I) 2025-26		
	Department: Chemical Engineering		Class: Final Year B. Tech.
	Subject: Piping Engineering	Subject Code: PEC-CH405	
Date: 09/10/2025	Time: 14:30 - 15:15		Max. Marks: 20

- Note: i. Attempt any two questions.
 ii. Assume suitable data if necessary
 iii. Use of non-programmable calculator is permitted

Q.N.	Questions		Marks	CO	BT
Q.1	a	Explain how changes in Reynolds number determine whether the flow is laminar, transitional, or turbulent.	04	1&2	2&3
	b	Describe the effects of pipe surface roughness on fluid flow.	04	3	2&3
	c	What do you understand by Apparent Reynolds number?	02	6	2
Q.2	a	What do you understand by minor losses in piping? Explain loss in a suddenly expanding pipe.	05	4	3&4
	b	Derive the equation for velocity distribution of laminar flow in pipe.	05	2	4
Q.3	a	Water, $\rho=998 \text{ kg/m}^3$, $\gamma = 1.005 \times 10^{-6} \text{ m}^2/\text{s}$ flows through a 6.0 cm tube diameter, 30 m long, at a flow rate of 0.34 lit/min. If the pipe discharges to the atmosphere, determine the supply pressure if the tube is inclined 10° above the horizontal in the flow direction.	08	2	5
	b	What do you understand by head loss?	02	2	1



Course Outcomes (CO)

- CO-1 Capable to understand the concepts of the nature of fluid flow
- CO-2 Ability to analyze Flow of liquid in Pipe.
- CO-3 Should understand the concept of Pipe surface roughness effects.
- CO-4 Get familiar with phenomenon of pipe networking.
- CO-5 Understand the difference between pipes and tubes.
- CO-6 Understand the Apparent Reynolds.

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Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Chemical Engineering
In-Semester Examination- II
Academic Year: 2025-26 (semester I)

Class: Final Year B. Tech	Course: Industrial Organization and Management	Course Code: HMC-CH404
Date: 10/10/2025	Time: 14:30 to 15:15	Max. Marks: 20

Note:

- Solve any *TWO* questions.
- Figures to right indicate full marks.

Q.N.			Marks	CO	BT
Q.1	a	Define the personnel management, also discuss the functions of the personnel management.	05	CO1	BT1
	b	What is human resource planning? Discuss the process of human resource planning.	05	CO1	BT1
Q.2	a	Explain the methods of job evaluation with its merits and demerits.	10	CO2	BT2
Q.3	a	Explain in brief the methods of wage payment.	05	CO2	BT1
	b	Discuss various method of employee training.	05	CO1	BT2

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	1. Understand the management process and structure in the industry so that it will help them to work in a better way.
CO 2	2. Develop an efficient methodology for industrial management.
CO 3	3. Cater the issues related to current industrial amendments.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Chemical Engineering
In-Semester Examination II
Academic Year: 2025-26 (Semester – I)

Class: Final Year B.Tech.	Subject: Transport Phenomena	Subject Code: PCC-CH403
Date: 11.10.2025	Time: 14:30 to 15:15	Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/Sharing of stationery, calculator etc. not allowed.
- iv. Figures to right indicate full marks.
- v. The symbols carry usual meaning.

Q.N.	Question	Marks	CO	BT
Q.1	Create a velocity distribution equation in flow through a circular tube using equation of motion. The equations of motion in terms of τ given on Page 2.	7	CO2	BT6
Q.2	The wall of an oven consists of three layers of bricks. The inside of the oven is made up of 0.2 m fire brick, surrounded by 0.1 m thickness of insulating brick and the outer side layer is made up of 0.15 m of building material. The oven operates at 870 °C and the outside temperature is 50 °C. Thermal conductivities of fire brick, insulating brick and building brick are 0.5, 0.225 and 0.6 kcal/h.m.°C respectively (a) How much heat is lost per m ² of the surface and (b) What are the temperatures at the interfaces of the layers.	8	CO1	BT5
Q.3	What is conservation of mass? Develop the continuity equation using mass balance over a volume element $\Delta x \Delta y \Delta z$.	5	CO4	BT 1 & 3

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Learn the fundamentals of transport phenomena
CO 2	Identify transport properties of molecular momentum, energy and mass transport
CO 3	Adopt the shell balance approach to transport processes
CO 4	Familiarize the equations of change
CO 5	Recognize analogies among momentum, heat and mass transfer

§B.5 THE EQUATION OF MOTION IN TERMS OF τ

$$[\rho D\mathbf{v}/Dt = -\nabla p - [\nabla \cdot \boldsymbol{\tau}] + \rho \mathbf{g}]$$

Cartesian coordinates (x, y, z) :^a

$$\rho \left(\frac{\partial v_x}{\partial t} + v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y} + v_z \frac{\partial v_x}{\partial z} \right) = -\frac{\partial p}{\partial x} - \left[\frac{\partial}{\partial x} \tau_{xx} + \frac{\partial}{\partial y} \tau_{yx} + \frac{\partial}{\partial z} \tau_{zx} \right] + \rho g_x \quad (\text{B.5-1})$$

$$\rho \left(\frac{\partial v_y}{\partial t} + v_x \frac{\partial v_y}{\partial x} + v_y \frac{\partial v_y}{\partial y} + v_z \frac{\partial v_y}{\partial z} \right) = -\frac{\partial p}{\partial y} - \left[\frac{\partial}{\partial x} \tau_{xy} + \frac{\partial}{\partial y} \tau_{yy} + \frac{\partial}{\partial z} \tau_{zy} \right] + \rho g_y \quad (\text{B.5-2})$$

$$\rho \left(\frac{\partial v_z}{\partial t} + v_x \frac{\partial v_z}{\partial x} + v_y \frac{\partial v_z}{\partial y} + v_z \frac{\partial v_z}{\partial z} \right) = -\frac{\partial p}{\partial z} - \left[\frac{\partial}{\partial x} \tau_{xz} + \frac{\partial}{\partial y} \tau_{yz} + \frac{\partial}{\partial z} \tau_{zz} \right] + \rho g_z \quad (\text{B.5-3})$$

^a These equations have been written without making the assumption that τ is symmetric. This means, for example, that when the usual assumption is made that the stress tensor is symmetric, τ_{xy} and τ_{yx} may be interchanged.

Cylindrical coordinates (r, θ, z) :^b

$$\rho \left(\frac{\partial v_r}{\partial t} + v_r \frac{\partial v_r}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_r}{\partial \theta} + v_z \frac{\partial v_r}{\partial z} - \frac{v_\theta^2}{r} \right) = -\frac{\partial p}{\partial r} - \left[\frac{1}{r} \frac{\partial}{\partial r} (r \tau_{rr}) + \frac{1}{r} \frac{\partial}{\partial \theta} \tau_{r\theta} + \frac{\partial}{\partial z} \tau_{rz} - \frac{\tau_{\theta\theta}}{r} \right] + \rho g_r \quad (\text{B.5-4})$$

$$\rho \left(\frac{\partial v_\theta}{\partial t} + v_r \frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\theta}{\partial \theta} + v_z \frac{\partial v_\theta}{\partial z} + \frac{v_r v_\theta}{r} \right) = -\frac{1}{r} \frac{\partial p}{\partial \theta} - \left[\frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \tau_{r\theta}) + \frac{1}{r} \frac{\partial}{\partial \theta} \tau_{\theta\theta} + \frac{\partial}{\partial z} \tau_{\theta z} + \frac{\tau_{r\theta} - \tau_{\theta r}}{r} \right] + \rho g_\theta \quad (\text{B.5-5})$$

$$\rho \left(\frac{\partial v_z}{\partial t} + v_r \frac{\partial v_z}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_z}{\partial \theta} + v_z \frac{\partial v_z}{\partial z} \right) = -\frac{\partial p}{\partial z} - \left[\frac{1}{r} \frac{\partial}{\partial r} (r \tau_{rz}) + \frac{1}{r} \frac{\partial}{\partial \theta} \tau_{\theta z} + \frac{\partial}{\partial z} \tau_{zz} \right] + \rho g_z \quad (\text{B.5-6})$$



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Department of Chemical Engineering
In-Semester Examination II
Academic Year: 2025-26 (semester I)

Class: Final Year	Course: Process Modelling & Simulation	Course Code: PCC-CH402
Date: 13.10.2025	Time: 14:30 to 15:15	Max. Marks: 20

Note:

- iii. Assume suitable data whenever necessary and state it clearly.
iv. Figures to the right indicate full marks.

Q.N.

	Question	Mark	CO	BT
Q.1	(a) Consider first order non-isothermal reaction is carried out in the batch reactor. You are required to provide detail process description, schematic, list of assumptions and model equations describing above the process	7	3	2
	OR			
	(b) Develop the state model that describes dynamic behaviour of the system consisting of two stirred tank heaters arranged like forming non-interacting system. The flowrates of the effluent streams are assumed to be proportional to liquid static pressure that causes flow of liquid. The cross-section areas of two tanks are A_1 and A_2 (m^3) respectively and flowrates are volumetric. No vapour is produced either first or second tank. List state variables and prove that system is mathematically consistent.	7	2	6
Q.2	(a) Define Murphree vapor phase efficiency of a tray for multicomponent non-ideal distillation column. Formulate all the model equations for this system. List suitable assumptions.	6	2	2 & 3
	OR			
	(a) Develop mathematical model equations for single stage continuous liquid – liquid extraction process with two components gets transferred.	6	2	5
Q.3	(a) What do you mean by distributed parameter model? Do the modelling of tubular plug flow reactor which used to carry out first order reaction.	7	3	1&5
	OR			
	(a) For LPG vaporizer, state all assumptions and develop model equations when it is used to purify the more volatile constituents. Take the consideration of vapor phase dynamics.	7	3	4

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

CH402.1 Able to apply fundamental laws & principles to formulate process models

CH402.2 Able to develop process model system for chemical processes.

CH402.3 Able to select proper modelling system and find simulated results for chemical processes.

CH402.4 Able to apply and solve chemical processes using commercial software packages.